

Лабораторная работа №10

1. Решить СЛАУ методом правой прогонки;
2. Решить СЛАУ методом левой прогонки;
3. Решить СЛАУ методом встречной прогонки.

Варианты заданий:

$$\text{№ 1} \quad \begin{cases} 5x_1 + 3x_2 = 11, \\ 2x_1 - 7x_2 + 3x_3 = -3, \\ 3x_2 + 6x_3 + x_4 = 28, \\ x_3 - 5x_4 + 2x_5 = -7, \\ x_4 - 5x_5 = -21. \end{cases} \quad x=(1,2,3,4,5).$$

$$\text{№ 2} \quad \begin{cases} 4x_1 + x_2 = 5, \\ x_1 + 4x_2 + x_3 = 6, \\ x_2 + 4x_3 + x_4 = 6, \\ x_3 + 4x_4 + x_5 = 6, \\ x_4 + 4x_5 + x_6 = 6, \\ x_5 + 4x_6 = 5. \end{cases} \quad x=(1,1,1,1,1,1).$$

$$\text{№ 3} \quad \begin{cases} -5x_1 + x_2 = -4, \\ x_1 + 6x_2 + 2x_3 = 9, \\ 2x_2 - 7x_3 + 4x_4 = -1, \\ x_3 + 9x_4 - 4x_5 = 6, \\ 2x_4 - 10x_5 + 6x_6 = -2, \\ x_5 + 7x_6 = 8. \end{cases} \quad x=(1,1,1,1,1,1).$$

$$\text{№ 4} \quad \begin{cases} 3x_1 - x_2 = 3, \\ x_1 + 5x_2 - 2x_3 = -3, \\ 3x_2 - 7x_3 + 4x_4 = -10, \\ 4x_3 + 9x_4 - 3x_5 = 14, \\ 2x_4 - 10x_5 + 5x_6 = -3, \\ -x_5 + 5x_6 = 4. \end{cases} \quad x=(1,0,2,1,1,1).$$

$$\text{№ 5} \quad \begin{cases} 2x_1 + x_2 = 6, \\ x_1 - 7x_2 + 2x_3 = -8, \\ 3x_2 - 6x_3 + 2x_4 = -2, \\ 2x_3 + 7x_4 - 3x_5 = 12, \\ x_4 - 8x_5 + 5x_6 = -4, \\ x_5 - 3x_6 = -4. \end{cases} \quad x=(2,2,2,2,2,2).$$

$$\text{№ 6} \quad \begin{cases} 3x_1 - x_2 = 1, \\ 2x_1 - 7x_2 + x_3 = -12, \\ x_2 - 5x_3 + 2x_4 = 4, \\ 3x_3 - 8x_4 + x_5 = -6, \\ x_4 + 6x_5 - 3x_6 = 10, \\ 2x_5 - 5x_6 = -1. \end{cases} \quad x=(1,2,0,1,2,1).$$

$$\text{№ 7} \quad \begin{cases} 2x_1 - x_2 = 1, \\ 3x_1 + 6x_2 - 2x_3 = 9, \\ x_2 - 5x_3 + 2x_4 = 3, \\ 4x_3 - 7x_4 + x_5 = -6, \\ x_4 + 5x_5 + 2x_6 = 8, \\ -2x_5 + 7x_6 = 5. \end{cases} \quad x = (1, 1, 0, 1, 1, 1).$$

$$\text{№ 8} \quad \begin{cases} 5x_1 + x_2 = 6, \\ x_1 + 5x_2 + x_3 = 7, \\ x_2 + 5x_3 + x_4 = 7, \\ x_3 + 5x_4 + x_5 = 7, \\ x_4 + 5x_5 + x_6 = 7, \\ x_5 + 5x_6 = 6. \end{cases} \quad x = (1, 1, 1, 1, 1, 1).$$

$$\text{№ 9} \quad \begin{cases} -3x_1 + x_2 = -2, \\ x_1 - 4x_2 + x_3 = -2, \\ x_2 - 4x_3 + x_4 = -2, \\ x_3 - 4x_4 + x_5 = -2, \\ x_4 - 4x_5 + x_6 = -2, \\ x_5 - 3x_6 = -2. \end{cases} \quad x = (1, 1, 1, 1, 1, 1).$$

$$\text{№ 10} \quad \begin{cases} 5x_1 - 2x_2 = 1, \\ x_1 + 7x_2 - x_3 = 14, \\ x_2 + 5x_3 - 2x_4 = 3, \\ x_3 + 6x_4 - x_5 = 12, \\ x_4 + 8x_5 - x_6 = 8, \\ x_5 - 2x_6 = -3. \end{cases} \quad x = (1, 2, 1, 2, 1, 2).$$

$$\text{№ 11} \quad \begin{cases} -4x_1 - x_2 = -9, \\ -x_1 - 5x_2 + 2x_3 = -3, \\ -x_2 - 5x_3 + 2x_4 = -9, \\ -x_3 - 5x_4 + 2x_5 = -3, \\ -x_4 - 5x_5 + 2x_6 = -9, \\ -x_5 - 5x_6 = -7. \end{cases} \quad x = (2, 1, 2, 1, 2, 1).$$

$$\text{№ 12} \begin{cases} 5x_1 - x_2 = -4, \\ -x_1 + 5x_2 - 2x_3 = -2, \\ -x_2 + 5x_3 - 2x_4 = -2, \\ -x_3 + 5x_4 - 2x_5 = -2, \\ -x_4 + 5x_5 - 2x_6 = -2, \\ -x_5 + 5x_6 = -4. \end{cases} \quad x=(-1,-1,-1,-1,-1,-1).$$

$$\text{№ 13} \begin{cases} 5x_1 - x_2 = 5, \\ x_1 - 5x_2 + 2x_3 = 3, \\ x_2 - 5x_3 + x_4 = -5, \\ x_3 - 5x_4 + x_5 = 2, \\ x_4 - 5x_5 + x_6 = -5, \\ x_5 - 5x_6 = 1. \end{cases} \quad x=(1,0,1,0,1,0).$$

$$\text{№ 14} \begin{cases} -3x_1 + x_2 = -1, \\ 2x_1 - 10x_2 + 2x_3 = -3, \\ 2x_2 - 10x_3 + 2x_4 = -3, \\ 2x_3 - 10x_4 + 2x_5 = -3, \\ 2x_4 - 10x_5 + 2x_6 = -3, \\ 2x_5 - 10x_6 = -4. \end{cases} \quad x=(0.5, 0.5, 0.5, 0.5, 0.5, 0.5).$$

$$\text{№ 15} \begin{cases} 4x_1 - 3x_2 = 1, \\ -3x_1 + 7x_2 - 3x_3 = 1, \\ -3x_2 + 7x_3 - 3x_4 = 1, \\ -3x_3 + 7x_4 - 3x_5 = 1, \\ -3x_4 + 7x_5 - 3x_6 = 1, \\ -3x_5 + 7x_6 = 4. \end{cases} \quad x=(1,1,1,1,1,1).$$

$$\text{№ 16} \begin{cases} 5x_1 - 2x_2 = 9, \\ -2x_1 + 5x_2 - 2x_3 = 3, \\ -2x_2 + 5x_3 - 2x_4 = 3, \\ -2x_3 + 5x_4 - 2x_5 = 3, \\ -2x_4 + 5x_5 - 2x_6 = 3, \\ -2x_5 + 5x_6 = 9. \end{cases} \quad x=(3,3,3,3,3,3).$$

$$\text{№ 17} \begin{cases} -10x_1 + 2x_2 = -6, \\ x_1 - 10x_2 + x_3 = -19, \\ 2x_2 - 10x_3 - x_4 = 2, \\ x_3 - 10x_4 - 2x_5 = -22, \\ x_4 - 10x_5 + 3x_6 = -8, \\ x_5 - 10x_6 = 1. \end{cases} \quad x=(1,2,0,2,1,0).$$

$$\text{№ 18} \begin{cases} 10x_1 - x_2 = 10, \\ x_1 - 10x_2 + 3x_3 = 4, \\ x_2 - 10x_3 - 2x_4 = -10, \\ 2x_3 - 10x_4 - x_5 = 1, \\ x_4 - 10x_5 + x_6 = -10, \\ 2x_5 - 10x_6 = 2. \end{cases} \quad x=(1,0,1,0,1,0).$$

$$\text{№ 19} \begin{cases} 5x_1 - 2x_2 = 1, \\ -2x_1 + 5x_2 - 2x_3 = 6, \\ -2x_2 + 5x_3 - 2x_4 = -3, \\ -2x_3 + 5x_4 - 2x_5 = 6, \\ -2x_4 + 5x_5 - 2x_6 = -3, \\ -2x_5 + 5x_6 = 8. \end{cases} \quad x=(1,2,1,2,1,2).$$

$$\text{№ 20} \begin{cases} 8x_1 - 2x_2 = 6, \\ -x_1 + 8x_2 - 2x_3 = 7, \\ -x_2 + 8x_3 - 2x_4 = -3, \\ -x_3 + 8x_4 - 2x_5 = 8, \\ -x_4 + 8x_5 - 2x_6 = -3, \\ -x_5 + 8x_6 = 8. \end{cases} \quad x=(1,1,0,1,0,1).$$

$$\text{№ 21} \begin{cases} -5x_1 - 2x_2 = 7, \\ -2x_1 + 7x_2 + 3x_3 = -8, \\ -2x_2 + 7x_3 - 3x_4 = -2, \\ -2x_3 + 7x_4 - 3x_5 = -2, \\ -2x_4 + 7x_5 - 3x_6 = -2, \\ -2x_5 + 7x_6 = -5. \end{cases} \quad x=(-1,-1,-1,-1,-1,-1).$$

$$\text{№ 22} \begin{cases} 3x_1 - x_2 = 4, \\ 3x_1 - 5x_2 + x_3 = 9, \\ 3x_2 - 5x_3 + x_4 = -1, \\ 3x_3 - 5x_4 + x_5 = 9, \\ 3x_4 - 5x_5 + x_6 = -1, \\ 3x_5 - 5x_6 = 8. \end{cases} \quad x = (1, -1, 1, -1, 1, -1).$$

$$\text{№ 23} \begin{cases} -4x_1 + 2x_2 = 6, \\ 2x_1 - 5x_2 + 2x_3 = -9, \\ x_2 - 5x_3 + 3x_4 = 9, \\ 3x_3 - 7x_4 + 3x_5 = -13, \\ x_4 - 6x_5 + 2x_6 = 9, \\ 2x_5 - 6x_6 = -8. \end{cases} \quad x = (-1, 1, -1, 1, -1, 1).$$

$$\text{№ 24} \begin{cases} 10x_1 + 5x_2 = 30, \\ 3x_1 - 10x_2 - 3x_3 = -20, \\ 4x_2 - 10x_3 + 3x_4 = -6, \\ 3x_3 - 10x_4 - 4x_5 = -22, \\ 2x_4 - 10x_5 - 3x_6 = -22, \\ x_5 - 10x_6 = -18. \end{cases} \quad x = (2, 2, 2, 2, 2, 2).$$

$$\text{№ 25} \begin{cases} 5x_1 - x_2 = 12, \\ 2x_1 - 12x_2 + 3x_3 = 34, \\ 3x_2 - 12x_3 - 3x_4 = -24, \\ -4x_3 - 12x_4 + 4x_5 = 24, \\ x_4 - 12x_5 + 2x_6 = -30, \\ -2x_5 - 12x_6 = 20. \end{cases} \quad x = (2, -2, 2, -2, 2, -2).$$